

Experiment 11

mongodb-11a.txt

```
C:\Users\Dell>mongod --version
```

```
db version v8.3.11
```

```
Build Info: {
  "version": "8.3.11",
  "gitVersion": "b8b83d6e6e1aed495578ad8c6f9234ad329c9955",
  "modules": [],
  "allocator": "tcmalloc-gperf",
  "environment": {
    "distmod": "windows",
    "distarch": "amd64",
    "target_arch": "amd64"
  }
}
```

```
C:\Users\Dell>mongosh --version
```

```
2.12.0
```

```
C:\Users\Dell>mongosh "mongodb+srv://cluster0.v24kq43.mongodb.net/" --username student
```

```
Enter password: *****
```

```
Current Mongosh Log ID: 6aba22e5b54db9e748008506
```

```
Connecting to:          mongodb+srv://<credentials>@cluster0.v24kq43.mongodb.net/?appName=mongosh+2.12.0
```

```
Using MongoDB:          8.0.32
```

```
Using Mongosh:          2.12.0
```

```
For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/
```

```
Atlas atlas-of3oig-shard-0 [primary] test>
```

```
Atlas atlas-of3oig-shard-0 [primary] test> db.version()
```

```
8.0.32
```

```
Atlas atlas-of3oig-shard-0 [primary] test> show dbs
```

```
college  72.00 KiB
```

```
library  72.00 KiB
```

```
admin      0 B
```

```
local      0 B
```

```
Atlas atlas-of3oig-shard-0 [primary] test> db.createCollection("bookscollection")
```

```
{ ok: 1 }
```

```
Atlas atlas-of3oig-shard-0 [primary] test> db.books.insertMany([
```

```
|   { title: "1984", author: "George Orwell", year: 1949, genre: "Dystopian", price: 12.99 },
```

```
|   { title: "To Kill a Mockingbird", author: "Harper Lee", year: 1960, genre: "Fiction", price: 14.99 },
```

```
|   { title: "The Great Gatsby", author: "F. Scott Fitzgerald", year: 1925, genre: "Classic", price: 10.99
```

```
},
```

```
|   { title: "Moby Dick", author: "Herman Melville", year: 1851, genre: "Adventure", price: 9.99 },
```

```
|   { title: "Brave New World", author: "Aldous Huxley", year: 1932, genre: "Dystopian", price: 11.99 }
```

```
| ])
```

```
|
```

```
{
```

```
  acknowledged: true,
```

```
  insertedIds: {
```

```
    '0': ObjectId('6aba260fb54db9e748008507'),
```

```
    '1': ObjectId('6aba260fb54db9e748008508'),
```

```
    '2': ObjectId('6aba260fb54db9e748008509'),
```

```
    '3': ObjectId('6aba260fb54db9e74800850a'),
```

```
    '4': ObjectId('6aba260fb54db9e74800850b')
```

```
  }
```

```
}
```

```
Atlas atlas-of3oig-shard-0 [primary] test> show collections
```

```
|
```

```
books
```

```
bookscollection
```

```
Atlas atlas-of3oig-shard-0 [primary] test> db.bookscollection.drop()
```

```
|
```

```
true
```

```
Atlas atlas-of3oig-shard-0 [primary] test> db.dropDatabase()
```

```
|  
{ ok: 1, dropped: 'test' }
```

mongodb-11b.txt

```
C:\Users\Dell>mongosh --version
2.12.0
```

```
C:\Users\Dell>mongosh
Current Mongosh Log ID: 6abdf59bbcc6acfcc4f1d1d1
Connecting to:      mongodb://127.0.0.1:27017/
?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB:      8.3.11
Using Mongosh:       2.12.0
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

```
-----
The server generated these startup warnings when booting
2026-09-24T15:12:28.557+05:30: Access control is not enabled for the database. Read and write access to
data and configuration is unrestricted
-----
```

```
test> use library
switched to db library
library> db
library
library> db.books.insertMany([
|   {
|     title: "The Great Gatsby",
|     author: "F. Scott Fitzgerald",
|     year: 1925,
|     genre: "Classic",
|     price: 450
|   },
|   {
|     title: "1984",
|     author: "George Orwell",
|     year: 1949,
|     genre: "Dystopian",
|     price: 500
|   },
|   {
|     title: "Animal Farm",
|     author: "George Orwell",
|     year: 1945,
|     genre: "Dystopian",
|     price: 350
|   },
|   {
|     title: "To Kill a Mockingbird",
|     author: "Harper Lee",
|     year: 1960,
|     genre: "Classic",
|     price: 550
|   },
|   {
|     title: "The Hobbit",
|     author: "J.R.R. Tolkien",
|     year: 1937,
|     genre: "Fantasy",
|     price: 600
|   },
|   {
|     title: "Harry Potter",
|     author: "J.K. Rowling",
|     year: 1997,
|     genre: "Fantasy",
|     price: 700
|   }
| ])
{
  acknowledged: true,
```

```

insertedIds: {
  '0': ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
  '1': ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
  '2': ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
  '3': ObjectId('6abdf5c3bcc6acfcc4f1d1d5'),
  '4': ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
  '5': ObjectId('6abdf5c3bcc6acfcc4f1d1d7')
}
}
library> db.books.find()
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,
    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    year: 1960,
    genre: 'Classic',
    price: 550
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d7'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 1997,
    genre: 'Fantasy',
    price: 700
  }
]
library> db.books.find().pretty()
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  },
  {

```

```

    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,
    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    year: 1960,
    genre: 'Classic',
    price: 550
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d7'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 1997,
    genre: 'Fantasy',
    price: 700
  }
]
library> db.books.find().sort({ year: 1 })
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,

```

```

    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    year: 1960,
    genre: 'Classic',
    price: 550
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d7'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 1997,
    genre: 'Fantasy',
    price: 700
  }
]
library> db.books.find().sort({ year: -1 })
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d7'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 1997,
    genre: 'Fantasy',
    price: 700
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    year: 1960,
    genre: 'Classic',
    price: 550
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,
    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  }
]

```

```

library> db.books.find().limit(3)
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,
    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  }
]
library> db.books.find().sort({ year: 1 }).limit(3)
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    year: 1925,
    genre: 'Classic',
    price: 450
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  }
]
library> db.books.find({ genre: "Dystopian" })
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d3'),
    title: '1984',
    author: 'George Orwell',
    year: 1949,
    genre: 'Dystopian',
    price: 500
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d4'),
    title: 'Animal Farm',
    author: 'George Orwell',
    year: 1945,
    genre: 'Dystopian',
    price: 350
  }
]

```

```

    }
  ]
library> db.books.find({ genre: "Fantasy" })
[
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d6'),
    title: 'The Hobbit',
    author: 'J.R.R. Tolkien',
    year: 1937,
    genre: 'Fantasy',
    price: 600
  },
  {
    _id: ObjectId('6abdf5c3bcc6acfcc4f1d1d7'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 1997,
    genre: 'Fantasy',
    price: 700
  }
]
library> db.books.createIndex({ author: 1 })
author_1
library> db.books.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { author: 1 }, name: 'author_1' }
]
library> db.books.aggregate([
  {
    $group: {
      _id: "$genre",
      count: { $sum: 1 }
    }
  }
])
[
  { _id: 'Fantasy', count: 2 },
  { _id: 'Classic', count: 2 },
  { _id: 'Dystopian', count: 2 }
]
library> db.books.aggregate([
  {
    $group: {
      _id: "$genre",
      count: { $sum: 1 }
    }
  },
  {
    $sort: {
      count: -1
    }
  }
])
[
  { _id: 'Dystopian', count: 2 },
  { _id: 'Classic', count: 2 },
  { _id: 'Fantasy', count: 2 }
]
library> db.books.aggregate([
  {
    $group: {
      _id: "$genre",
      avgPrice: { $avg: "$price" }
    }
  }
])
[
  { _id: 'Dystopian', avgPrice: 425 },
  { _id: 'Classic', avgPrice: 500 },
  { _id: 'Fantasy', avgPrice: 650 }
]

```

```

]
library> db.books.aggregate([
|   {
|     $group: {
|       _id: "$genre",
|       avgPrice: { $avg: "$price" }
|     },
|     {
|       $sort: {
|         avgPrice: -1
|       }
|     }
|   ]
| )
[
  { _id: 'Fantasy', avgPrice: 650 },
  { _id: 'Classic', avgPrice: 500 },
  { _id: 'Dystopian', avgPrice: 425 }
]
library> db.books.aggregate([
|   {
|     $match: {
|       year: { $gt: 1900 }
|     },
|     {
|       $group: {
|         _id: "$genre",
|         count: { $sum: 1 },
|         avgPrice: { $avg: "$price" }
|       },
|       {
|         $sort: {
|           count: -1
|         }
|       },
|       {
|         $project: {
|           genre: "$_id",
|           count: 1,
|           avgPrice: 1,
|           _id: 0
|         }
|       }
|     }
|   ]
| )
[
  { count: 2, avgPrice: 425, genre: 'Dystopian' },
  { count: 2, avgPrice: 500, genre: 'Classic' },
  { count: 2, avgPrice: 650, genre: 'Fantasy' }
]
library>

```